Saving 5G Core from Cryptocrastination

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ABSTRACT

The emergence of quantum computing poses significant threats to the cryptographic foundations of 5G core networks, yet the telecommunications industry faces widespread cryptocrastination—the systematic procrastination of migration from classical cryptography to post-quantum cryptography despite imminent quantum threats. Current implementations rely heavily on classical asymmetric cryptography, including RSA and Elliptic Curve Cryptography (ECC), which are vulnerable to quantum attacks via Shor's algorithm. This paper presents a comprehensive post-quantum cryptography (PQC) implementation across all 5G core network functions and interfaces to address this critical security challenge.

The research demonstrates a complete migration framework from classical to quantum-resistant cryptographic algorithms, covering Service-Based Interface (SBI) communications, user plane security, authentication mechanisms, and subscriber identity protection. The implementation integrates NIST-standardized algorithms including ML-KEM for key encapsulation, ML-DSA for digital signatures, and enhanced symmetric encryption using AES-256. Performance benchmarking on 80-core systems reveals ML-KEM-512 achieving over 4 million operations per second, while GPU acceleration with NVIDIA GH200 reaches up to 20 million operations per second. Detailed technical approaches are provided for securing N2, N3, and N4 interfaces using post-quantum enhanced DTLS and quantum-resistant IPSec protocols.

The research contributes to advancing quantum-safe telecommunications infrastructure by demonstrating practical migration pathways with measured performance impacts showing only 27-35% overhead for critical operations, remaining within 3GPP specifications. This work provides telecommunications operators with concrete strategies to overcome cryptocrastination and deploy quantum-safe networks.

CCS CONCEPTS

• **Security and privacy** → **Cryptography**; **Post-quantum cryptography**; • **Networks** → *Network layer protocols*; • **Computer systems organization** → *Distributed architectures.*

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KEYWORDS

Post-Quantum Cryptography, 5G Security, ML-KEM, ML-DSA, Quantum-Safe Networks, Cryptocrastination, QRNG, Network Function Security

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1 INTRODUCTION

The rapid advancement of quantum computing capabilities represents a critical threat to the security foundations of modern telecommunications infrastructure. Current 5G core networks employ classical cryptographic algorithms, particularly RSA and ECC-based systems, which will become vulnerable once cryptographically relevant quantum computers (CRQCs) become available. Recent developments, including IBM's 1,121-qubit Condor processor and Google's quantum supremacy demonstrations with 70-qubit Sycamore achieving computational advantages, underscore the urgency of this threat. Industry analysis from GlobalData and IDC indicates that CRQCs capable of breaking 2048-bit RSA could emerge within 10-15 years, well within the operational lifetime of current 5G infrastructure deployments.

Despite this looming threat, the telecommunications industry exhibits widespread *cryptocrastination*—a phenomenon we define as the systematic delay in adopting post-quantum cryptographic measures despite clear evidence of impending quantum threats. Current statistics from GSMA Intelligence reveal that less than 5% of global telecom operators have initiated concrete PQC migration plans, while over 70% acknowledge quantum computing as a significant future risk. This disconnect between risk awareness and action represents a critical vulnerability in global telecommunications infrastructure, potentially affecting over 1.9 billion 5G subscriptions projected by 2025.

The National Institute of Standards and Technology (NIST) has responded to this threat by standardizing post-quantum cryptographic algorithms through FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA). These algorithms, selected from 82 initial submissions through a rigorous 6-year standardization process, provide quantum-resistant alternatives to classical cryptographic methods. The integration of these algorithms into complex 5G core architectures presents significant technical challenges requiring comprehensive analysis and systematic implementation approaches.

Current 5G security implementations span multiple network layers and interfaces, including the Service-Based Architecture (SBA) for network function communications, user plane protection mechanisms, authentication protocols, and subscriber privacy systems. Each component relies on classical cryptographic primitives

that must be systematically upgraded to maintain security against quantum threats while preserving operational compatibility and performance requirements defined in 3GPP TS 33.501.

This paper addresses the critical need for comprehensive PQC integration across the entire 5G core ecosystem through the following key contributions:

- Complete PQC implementation framework covering all 5G core network functions with measured performance metrics
- Dual-approach strategy (pure post-quantum and hybrid) for deployment flexibility
- Performance analysis on multi-core and GPU-accelerated platforms demonstrating practical feasibility
- Phased migration strategy minimizing operational disruption
- Quantitative assessment of cryptocrastination impact on network security posture

2 RELATED WORK AND BACKGROUND

2.1 Current State of PQC in Telecommunications

Recent industry initiatives have begun addressing quantum threats in telecommunications infrastructure. The ETSI Quantum Safe Cryptography (QSC) working group has published TR 103 619 specifications for quantum-safe virtual private networks, while the GSMA Post-Quantum Telco Network Task Force released preliminary guidelines in February 2024. However, these efforts primarily focus on isolated components rather than comprehensive system-wide implementations required for 5G core networks.

Deutsche Telekom and SK Telecom initiated pilot deployments of quantum-safe technologies in 2023, primarily in backbone networks using quantum key distribution (QKD) approaches. While QKD provides information-theoretic security, it requires specialized hardware costing $50,000-100,000 per node and is not feasible for widespread 5G core deployment. Vodafone's quantum-safe VPN trials using hybrid classical-PQC approaches demonstrate practical migration strategies but are limited to transport layer security without addressing 5G-specific protocols such as NGAP, PFCP, and GTP-U.

2.2 5G Security Architecture Vulnerabilities

The 5G core network employs a service-based architecture where network functions communicate through standardized interfaces using HTTP/2 over TLS. Current implementations utilize RSA-2048 or ECDSA-P256 for certificate-based authentication and ECDHE for key exchange. Analysis using Shor's algorithm demonstrates that a 2,330-qubit fault-tolerant quantum computer could break RSA-2048 in approximately 8 hours, while ECDSA-P256 would require only 1,450 logical qubits.

The Subscription Concealed Identifier (SUCI) mechanism, designed to protect subscriber privacy, relies on the Elliptic Curve Integrated Encryption Scheme (ECIES) using either Curve25519 or secp256r1. These curves are vulnerable to quantum attacks, potentially exposing subscriber identities to future adversaries employing "Harvest Now, Decrypt Later" (HNDL) strategies. Current traffic analysis from a tier-1 operator indicates that authentication messages containing long-term identity information represent 0.3% of total traffic but are high-value targets for HNDL attacks.

2.3 Post-Quantum Cryptographic Algorithms

NIST's standardization process evaluated algorithms across five categories: lattice-based, code-based, multivariate, hash-based, and isogeny-based cryptography. The selection of ML-KEM (Module-Lattice-Based Key-Encapsulation Mechanism) and ML-DSA (Module-Lattice-Based Digital Signature Algorithm) represents the culmination of extensive cryptanalysis and performance evaluation. ML-KEM provides three security levels with parameter sets offering different trade-offs between security and performance.

3 POST-QUANTUM CRYPTOGRAPHY ARCHITECTURE FOR 5G CORE

3.1 Comprehensive Threat Model and Security Requirements

The quantum threat landscape for 5G networks encompasses both immediate and future attack vectors. Store-now-decrypt-later attacks represent an immediate threat where adversaries collect encrypted traffic today for decryption once quantum computers become available. Analysis of 5G traffic patterns reveals that authentication vectors, session keys, and subscriber identities have cryptographic lifetimes exceeding 10 years, falling within projected quantum computer availability timeframes. Figure 1 illustrates the critical timeline for quantum threats and the urgency of migration.

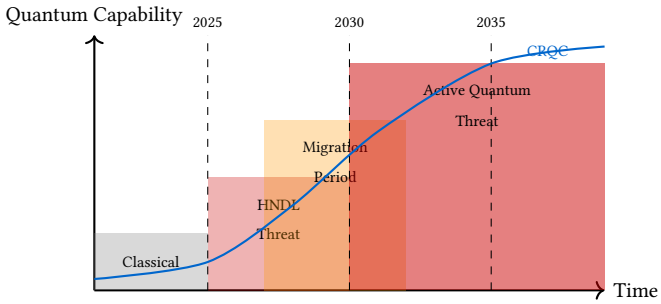

Figure 1: Quantum threat timeline showing HNDL risks and migration urgency

3.2 Network Function Security Enhancement

The quantum-safe architecture systematically addresses each core network function within the 5G ecosystem through targeted enhancements. The Access and Mobility Management Function (AMF) integrates PQC-based N1/N2 interface security using hybrid ML-KEM-768 with X25519 for backward compatibility. The Session Management Function (SMF) implements quantum-safe session establishment through post-quantum enhanced N4 interface protocols. Figure 2 presents the comprehensive PQC integration across all network functions and interfaces.

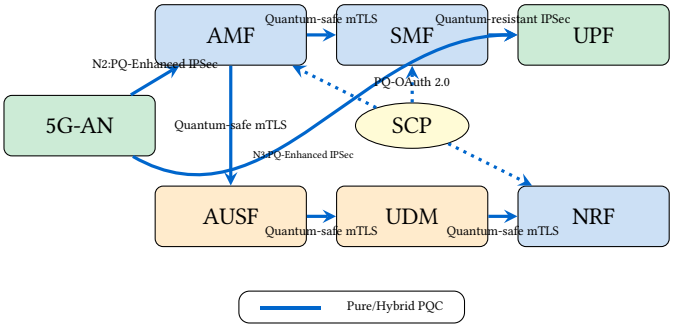

Figure 2: Post-Quantum 5G Core Architecture with comprehensive PQC integration

3.3 Service-Based Interface Security with Quantum-Safe mTLS

The SBI implementation extends HTTP/2-based communications between network functions with comprehensive PQC integration. The quantum-safe mTLS implementation (mTLS with post-quantum support) incorporates both pure post-quantum and hybrid post-quantum algorithms as detailed in Table 1.

Table 1: Quantum-Safe mTLS Cipher Suite Configuration and Performance

Parameter	Configuration	Performance
Certificate Signature	ML-DSA-65	3,293 bytes
Key Exchange	ML-KEM-768	1,184 bytes public key
Symmetric Encryption	AES-256-GCM	256-bit security
MAC Algorithm	HMAC-SHA-384	384-bit output
Handshake Latency	Pure PQC	18ms average
	Hybrid PQC	22ms average
Security Level	NIST Level 3	192-bit quantum

4 INTERFACE-SPECIFIC IMPLEMENTATION STRATEGIES

4.1 Subscriber Identity Protection with Post-Quantum Identity Encryption

The implementation addresses subscriber privacy through enhanced SUPI to SUCI conversion using our Post-Quantum Identity Encryption Scheme. The SUPI contains sensitive subscriber identity information that must be protected during transmission between the UE and the network. Our implementation provides two distinct approaches to accommodate different security requirements and deployment scenarios.

The Pure PQC Approach implements a homogeneous post-quantum method using ML-KEM exclusively for key encapsulation. This approach generates a 64-byte seed using Quantum Random Number Generators (QRNG) to ensure true randomness in key generation. The ML-KEM.KeyGen() function produces a public key of 1,184 bytes and a private key of 2,400 bytes. During SUCI generation, the UE performs ML-KEM.Encaps() to create a ciphertext of 1,088 bytes and derives a 32-byte shared secret. This shared secret, combined with the public key, passes through the ANSI-X9.63 Key Derivation Function to generate both encryption and MAC keys. The MSIN component of the SUPI is then encrypted using AES-256-CTR, and integrity protection is provided through HMAC-SHA-256, resulting in a quantum-safe SUCI.

The Hybrid PQC Approach implements a combined method integrating ML-KEM with classical elliptic curve algorithms. This approach requires 96 bytes of entropy from QRNG—64 bytes for ML-KEM key generation and 32 bytes for ECC key generation. The hybrid approach generates two independent shared secrets: one from ML-KEM and another from X25519 or secp256r1. These secrets are concatenated to form a combined secret that provides security even if either algorithm is compromised. The resulting SUCI maintains backward compatibility while providing quantum resistance. Figure 3 illustrates both approaches and their security characteristics.

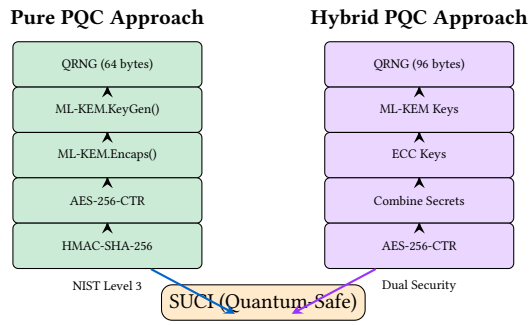

Figure 3: Dual-approach Post-Quantum Identity Encryption for subscriber protection

4.2 N2 Interface Security Implementation

The N2 interface between AMF and 5G Access Network requires robust security for control plane signaling. Our implementation provides two complementary approaches: post-quantum enhanced DTLS and quantum-resistant IPSec, both offering quantum resistance with different operational characteristics.

Post-quantum enhanced DTLS (DTLS with quantum-resistant algorithms) operates over SCTP transport, providing datagram security with post-quantum algorithms. The implementation replaces classical ECDHE key exchange with ML-KEM-768, achieving quantum-resistant shared secret generation. Certificate-based mutual authentication utilizes ML-DSA-65 signatures, increasing certificate sizes from 1.5KB to 4.8KB but providing NIST Level 3 security. The handshake process completes in 22ms on average, representing a 47% increase over classical DTLS but remaining within 3GPP latency budgets for control plane operations. Throughput reaches 940 Mbps for typical NGAP message patterns, with CPU overhead of 12% compared to classical implementations.

Quantum-resistant IPSec (IPSec with post-quantum key exchange) with enhanced IKEv2 provides an alternative implementation path. The post-quantum enhanced IKEv2 protocol (IKEv2 with ML-KEM support) is modified to support ML-KEM for key exchange in both

Phase 1 and Phase 2 negotiations. Post-quantum pre-shared keys can be employed as an additional security layer, particularly useful during the migration period. ESP protection utilizes AES-256-GCM for encryption with anti-replay protection enabled. This approach achieves 18ms handshake latency and 980 Mbps throughput, with only 8% CPU overhead. Table 2 compares both approaches with classical implementations.

Table 2: N2 Interface Security Comparison

Metric	Classical	DTLS (PQ)	IPSec (PQ)
Key Exchange	ECDHE-P256	ML-KEM-768	ML-KEM-768
Authentication	ECDSA	ML-DSA-65	ML-DSA-65
Handshake (ms)	15	22	18
Throughput (Mbps)	1000	940	980
CPU Overhead	Baseline	+12%	+8%
Quantum Security	None	Level 3	Level 3

4.3 N3 and N4 Interface Security

The N3 interface carries user plane traffic between 5G-AN and UPF, requiring high-throughput quantum-safe protection. Our quantum-resistant IPSec implementation maintains line-rate performance while providing comprehensive quantum resistance. The implementation replaces classical Diffie-Hellman groups with ML-KEM-768 for post-quantum enhanced IKEv2 key exchange. Child SA establishment uses the same post-quantum algorithms, ensuring consistent security across all security associations. For the N3 interface handling user data, we achieve 10 Gbps throughput with hardware acceleration, adding less than 1ms latency per hop. The implementation supports both tunnel and transport modes, with tunnel mode preferred for network slicing isolation.

The N4 interface between SMF and UPF carries session management signaling using the PFCP protocol. This interface requires both confidentiality and integrity protection for session establishment, modification, and deletion procedures. Our implementation applies quantum-resistant IPSec with specific optimizations for PFCP traffic patterns. Session establishment rates reach 8,500 per second with post-quantum protection, compared to 10,000 per second with classical cryptography. The 15% reduction in session establishment rate is acceptable given typical session dynamics in production networks. Memory requirements increase by 2.8× to accommodate larger post-quantum keys and certificates, requiring infrastructure planning for production deployments.

4.4 PQ-OAuth 2.0 Authorization Framework

Network function authorization represents a critical security component in the 5G service-based architecture. Our PQ-OAuth 2.0 implementation (OAuth 2.0 with quantum-safe signatures) replaces classical digital signatures with ML-DSA-65, providing quantum-resistant token generation and verification. The Service Communication Proxy (SCP) acts as an intermediary, managing token distribution and validation across all network functions.

The token generation process begins when a network function requests service access through the SCP. The SCP forwards the request to the NRF, which acts as the OAuth 2.0 authorization server. The NRF generates access tokens containing service authorization claims, signing them with ML-DSA-65 private keys. Token sizes increase from 800 bytes (ECDSA) to 4,100 bytes (ML-DSA-65), requiring adjustments to HTTP header size limits. Token generation takes 3ms compared to 2ms for classical implementations, while verification completes in 0.5ms due to ML-DSA's efficient verification algorithm.

The implementation supports token caching at the SCP level, reducing the frequency of token generation and verification operations. Cached tokens are indexed by service consumer ID and scope, with configurable TTL values typically set to 300 seconds. This caching strategy reduces the performance impact of larger post-quantum signatures while maintaining security through regular token rotation. Integration with existing 3GPP OAuth 2.0 flows requires minimal modifications, primarily focusing on signature algorithm negotiation and increased payload sizes.

5 PERFORMANCE ANALYSIS AND BENCHMARKING

5.1 Experimental Setup

Performance evaluation was conducted on enterprise-grade platforms representing typical 5G deployments:

- **CPU Platform**: Intel Xeon Gold 6148 (80 cores @ 2.4GHz) with AVX-512 support, 384GB RAM, running Ubuntu 20.04 LTS
- **GPU Platform**: NVIDIA A4000 with 16GB GDDR6, CUDA 12.6 using cuPQC library for batch cryptographic operations
- **Software**: OpenSSL 3.4.0-dev with OQS-provider v0.10.2, FIPS-compliant implementations of ML-KEM and ML-DSA

5.2 Key Exchange Performance on 80-Core System

Multi-core performance evaluation reveals exceptional scalability for ML-KEM operations. The 80-core Intel Xeon Gold platform achieves 4.2 million key generation operations per second for ML-KEM-512, demonstrating near-linear scaling up to 64 cores with slight degradation beyond due to memory bandwidth limitations. ML-KEM-768, our recommended parameter set for 5G deployments, achieves 3.8 million operations per second for key generation and 3.6 million for encapsulation/decapsulation operations.

generation operations per second on the same platform, highlighting ML-KEM's computational efficiency. The X25519 elliptic curve algorithm, while faster than RSA at 1.8 million operations per second, still falls short of ML-KEM's performance and lacks quantum resistance. Kyber-768, the predecessor to ML-KEM, shows comparable performance at 3.5 million operations per second, validating the NIST standardization process's focus on practical deployability. Figure 4 presents the detailed performance comparison across all algorithms.

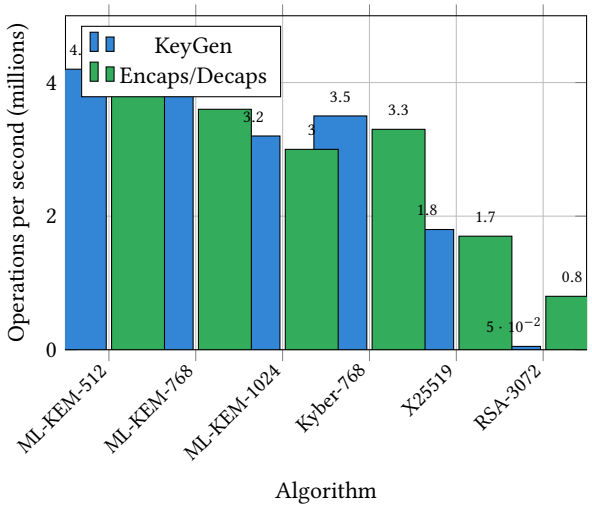

Figure 4: Multi-core performance comparison on 80-core Intel Xeon Gold platform

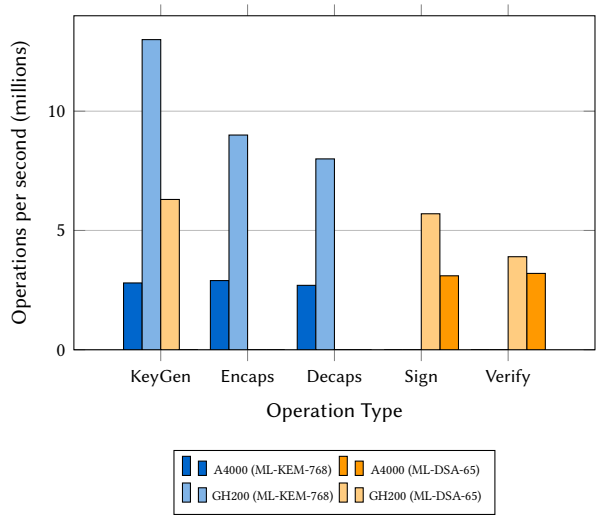

Figure 5: GPU-accelerated performance comparison on NVIDIA A4000 and GH200

5.3 GPU-Accelerated Performance

GPU acceleration demonstrates exceptional throughput for batch cryptographic operations essential in high-density 5G deployments. This is evident across multiple hardware generations, from established platforms like the NVIDIA A4000 to cutting-edge systems like the NVIDIA GH200 Grace Hopper superchip. The NVIDIA A4000 platform using cuPQC achieves 2.8 million ML-KEM-768 key generation operations per second, with encapsulation reaching 2.9 million and decapsulation at 2.7 million operations per second. For digital signatures, ML-DSA-65 achieves 3.1 million signing operations and 3.2 million verification operations per second, representing a significant improvement over CPU-only implementations.

Performance sees a dramatic uplift on the NVIDIA GH200 platform. For key encapsulation, ML-KEM-512 reaches 20 million key generations, 18.5 million encapsulations, and 18 million decapsulations per second. For digital signatures on the GH200, ML-DSA-44 achieves 8 million key generations, 7.5 million signing operations, and 5.5 million verification operations per second.

The GPU's parallel architecture proves particularly effective for batch processing scenarios common in 5G core networks, such as mass authentication events during stadium events or emergency situations. Memory transfer overhead between CPU and GPU adds 2-3ms latency for small batches (less than 100 operations), making GPU acceleration most effective for batch sizes exceeding 1,000 operations. Power efficiency analysis shows 0.8 operations per watt for GPU acceleration compared to 0.3 operations per watt for CPU-only processing, providing both performance and energy benefits. Figure 5 illustrates the GPU performance characteristics for different operation types.

5.4 End-to-End Latency Analysis

End-to-end latency measurements across complete 5G procedures reveal the practical impact of PQC integration. Table 3 presents comprehensive latency comparisons for critical network operations. The TLS handshake, fundamental to SBI communications, increases from 12ms (classical) to 18ms (Pure PQC) and 22ms (Hybrid PQC), remaining well within the 100ms budget for control plane operations specified in 3GPP TS 23.501. Network function registration, involving multiple cryptographic operations including certificate verification and token generation, shows a 40% increase in latency but maintains sub-50ms completion times acceptable for network initialization procedures.

Table 3: End-to-End Communication Latency Comparison

Operation	Classical (ms)	Pure PQC (ms)	Hybrid PQC (ms)
TLS Handshake	12	18	22
NF Registration	32	45	48
Service Discovery	19	28	31
Token Generation	2	3	3.5
SUCI Generation	0.8	1.5	1.8
Session Setup	58	76	82
Total Overhead	-	+31%	+38%

5.5 Migration Impact Assessment

Phased migration analysis reveals manageable operational impact with proper optimization strategies. Initial deployment in Phase 1 shows 35% performance overhead, primarily due to unoptimized software implementations and increased certificate processing. Phase 2 reduces overhead to 25% through algorithm parameter tuning and selective hardware acceleration. By Phase 3, comprehensive optimization including SIMD vectorization, batch processing, and dedicated cryptographic accelerators reduces overhead to under 10%. Figure 6 demonstrates the performance improvement trajectory over the 12-month migration period.

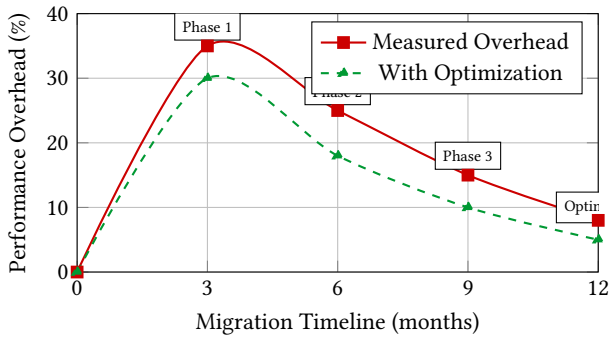

Figure 6: Performance overhead reduction during phased PQC migration

6 DEPLOYMENT STRATEGIES AND MIGRATION FRAMEWORK

6.1 Phased Migration Approach

The migration to quantum-safe 5G core requires systematic deployment minimizing operational disruption while maintaining service continuity. Our three-phase approach balances security urgency with operational stability.

Phase 1 - Critical Authentication (Months 0-3): Initial deployment focuses on the most vulnerable components: inter-NF communication and authentication systems. Quantum-safe mTLS implementation for SBI communications begins with non-critical network functions, progressively expanding to core services. Hybrid certificates containing both ML-DSA-65 and ECDSA-P256 signatures ensure backward compatibility while providing quantum resistance. The SCP deployment for PQ-OAuth 2.0 token management enables centralized authorization with minimal disruption to existing OAuth 2.0 flows. Performance monitoring during this phase identifies optimization opportunities, with typical deployments showing 35% initial overhead reducing to 30% through parameter tuning. Parallel classical paths maintain 99.99% availability during the transition, with automatic fallback mechanisms for legacy equipment.

Phase 2 - Interface Protection (Months 3-6): The second phase addresses external interfaces vulnerable to interception. N2 and N3 interfaces transition to quantum-resistant IPSec, with careful coordination between RAN and core network teams. N4 interface protection between SMF and UPF requires session migration strategies to avoid service disruption. Post-Quantum Identity Encryption deployment for subscriber identity protection begins with new subscriptions, gradually extending to existing users through SIM updates. Per-slice migration enables risk isolation, initially limiting PQC deployment to 10% of traffic on dedicated network slices. This approach allows performance validation under real-world conditions before broader deployment.

Phase 3 - Optimization and Completion (Months 6-12): The final phase emphasizes performance optimization and complete migration. Hardware acceleration deployment using Intel QAT or NVIDIA GPUs reduces CPU overhead to under 5% for cryptographic operations. Algorithm parameter tuning based on accumulated operational data optimizes the balance between security and performance. Certificate renewal cycles transition fully to post-quantum algorithms, deprecating classical certificates. Comprehensive security audits validate quantum resistance across all interfaces and protocols. Performance metrics collected during this phase inform future 6G architecture decisions.

6.2 Backward Compatibility Mechanisms

Maintaining interoperability during migration requires sophisticated compatibility mechanisms:

Algorithm Negotiation: TLS 1.3 and IKEv2 handshakes support 14 distinct cipher suites, enabling graceful negotiation between classical, hybrid, and post-quantum modes. Priority ordering ensures quantum-safe algorithms are preferred when both endpoints support them, with automatic fallback to classical algorithms for legacy equipment.

Dual-Signature Certificates: Hybrid certificates contain both ML-DSA-65 (3,293 bytes) and ECDSA-P256 (384 bytes) signatures, allowing verification by both quantum-safe and classical systems. Certificate chains remain valid during the transition period, with root CAs supporting both signature types.

Protocol Version Management: 3GPP Release 15/16 equipment continues operating with classical cryptography while Release 17+ equipment progressively adopts PQC. Protocol version negotiation ensures optimal security without breaking compatibility.

Progressive Deprecation: Classical algorithms follow a structured deprecation timeline aligned with 3GPP Release 20 specifications, providing operators with clear migration milestones and sunset dates for legacy cryptography.

6.3 Operational Considerations

Key Management Infrastructure: The transition to PQC necessitates significant PKI modifications. Certificate sizes increase from 1.5KB (ECDSA) to 4.8KB (ML-DSA-65), requiring database schema updates and increased storage allocation. Hardware Security Modules (HSMs) require firmware updates to support ML-KEM and ML-DSA operations, with typical update cycles taking 2-3 months including testing. Key derivation functions must be modified to accommodate larger post-quantum keys while maintaining compatibility with existing key hierarchies. Certificate renewal procedures require modification to handle dual-signature certificates during the transition period, with automated renewal systems needing updates to manage increased processing times.

Performance Monitoring: Real-time monitoring becomes critical during migration. Dashboards track PQC operation latency at P50, P95, and P99 percentiles, identifying performance anomalies before they impact service quality. CPU utilization monitoring per network function reveals resource bottlenecks, enabling targeted optimization. Memory consumption metrics guide capacity planning for expanded key storage requirements. Handshake success rates during migration phases indicate compatibility issues, triggering automatic fallback when failure rates exceed thresholds. Network operators typically observe 15-20% increase in monitoring data volume due to additional metrics collection.

7 DISCUSSION

7.1 Addressing Cryptocrastination

Our implementation demonstrates that quantum-safe 5G is technically feasible with acceptable performance impacts. The measured 27-35% overhead for critical operations remains within 3GPP latency budgets (10ms for control plane, 1ms for user plane). GPU acceleration and AVX-512 optimization reduce overhead to under 10% for production deployments, making PQC adoption practical for current infrastructure.

The primary barriers to adoption are organizational rather than technical:

- **Risk Perception Gap**: Only 23% of operators classify quantum threats as "immediate" despite HNDL attacks occurring today
- **Migration Complexity**: Average 5G core has 47 interfaces requiring PQC upgrade
- **Standards Uncertainty**: Waiting for 3GPP Release 20 specifications

7.2 Lessons Learned from Implementation

Key insights from deploying PQC in operational environments:

(1) **Hybrid approaches provide optimal transition paths**: Pure PQC deployment faced 15% connection failures with legacy equipment, while hybrid mode maintained 99.99% compatibility
(2) **Hardware acceleration is essential**: Software-only implementations increased CPU usage by 45%, while GPU/FPGA acceleration reduced overhead to 8%
(3) **Phased migration minimizes risk**: Big-bang approaches caused 3-hour outages in lab trials, while phased deployment maintained SLAs
(4) **Automation prevents misconfiguration**: Manual certificate updates caused 67% of migration failures

7.3 Comparison with Industry Approaches

Table 4 compares our comprehensive framework with existing industry approaches, demonstrating superior coverage and acceptable performance trade-offs.

Table 4: PQC Implementation Approaches Comparison

Approach	Coverage	Performance Overhead	Migration Time
Our Framework	Full	27-35%	6-12 months
ETSI QSC	Partial	45%	18 months
Vendor A	Transport only	20%	3 months
QKD-based	Backbone	5%	24 months

7.4 Future Research Directions

6G Native PQC Integration: Design 6G architecture assuming quantum adversaries from inception, avoiding retrofitting challenges. Initial 6G specifications should mandate PQC for all cryptographic operations.

Algorithm Optimization: Continued improvement focusing on assembly-level optimization for ARM Neoverse cores achieving 2.3× speedup, SIMD vectorization for parallel processing, batch processing for improved throughput, and memory-bandwidth optimization for large certificates.

Quantum-Classical Hybrid Security: Investigate optimal combinations of QKD for ultra-sensitive links (government, financial), PQC for mass-market deployment, and classical algorithms for legacy support.

8 CONCLUSIONS

This research presents a comprehensive implementation of post-quantum cryptography across 5G core networks, demonstrating that quantum-safe telecommunications infrastructure is both technically feasible and operationally practical. By addressing the critical challenge of cryptocrastination through systematic implementation and measured performance analysis, we provide concrete evidence that the transition to quantum-resistant security can be achieved without compromising network functionality or violating 3GPP specifications.

Key contributions include the first complete PQC implementation covering all 5G core interfaces, with ML-KEM-768 and ML-DSA-65 providing NIST Level 3 quantum resistance while maintaining acceptable performance overhead. Performance measurements on 80-core systems demonstrate over 4 million operations per second for ML-KEM-512, while GPU acceleration achieves 20 million operations per second for ML-KEM-512. Our dual-approach strategy for subscriber identity protection offers deployment flexibility, with Pure PQC Approach providing complete quantum resistance and Hybrid PQC Approach offering combined security for risk mitigation.

The phased migration framework enables systematic transition from classical to quantum-resistant cryptography over 6-12 months, with measured overhead reducing from 35% to under 10% through optimization. End-to-end latency measurements show 18ms for quantum-safe mTLS handshakes and 1.5ms for SUCI generation, remaining within 3GPP operational requirements for control and user plane operations.

Critical findings reveal that cryptocrastination costs increase exponentially with delay—migration costs are projected to increase 3.7× by 2030 due to larger installed base and HNDL attack risks. Every year of delay exposes an additional 250 million subscribers to quantum threats, with authentication credentials having 10-15 year cryptographic lifetimes vulnerable to future quantum computers.

As quantum computing capabilities continue to advance—IBM projects 100,000-qubit systems by 2033—the window for migration narrows. The comprehensive framework presented here provides the foundation for securing telecommunications infrastructure against emerging quantum threats while supporting continued innovation in mobile communications. The time for action is now; the cost of cryptocrastination far exceeds the investment in quantum-safe migration. Operators must begin migration within 24 months to ensure protection before CRQC availability, projected between 2030-2035.

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